

**B.E. Electrical Engineering Second Year Second
Semester Examination – 2023**

**Electrical Instrumentation
PART–II Comprehensive Analytical
Solutions**

Academic Article Style Notes with Exam-Oriented Derivations, Definitions
and Clean Diagrams

Diagram discipline followed in this solution:

- All circuit and block diagrams are drawn using straight or perpendicular connections as far as possible.
- Labels are placed away from wires and components so that words do not collapse on the diagram.
- Longer labels are split vertically, or placed above/below/left/right with spacing.
- Large diagrams are represented as clean block diagrams instead of crowded circuit sketches.

Prepared for semester-exam style writing: from grassroot idea to final
boxed result.

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Final Revision Summary

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Part-II Question List Covered

This document solves every Part-II question of the 2023 Electrical Instrumentation paper, including the OR alternatives. The representation is kept in a semester-answer style: first the physical idea, then definitions, then equations, then derivation, then final result.

1. CRO and Lissajous pattern, matched probe, delay line; OR: X-Y mode and time-base generator.
2. State-variable universal active filter and PLL linear model; PLL frequency synthesizer.
3. DAC/ADC numerical and theory questions; OR: dual-slope ADC.
4. Butterworth and Chebyshev filter theory; band-pass/band-stop design; OR: normalized Butterworth LPF design.

1 Solution to Question 1(a): Lissajous Frequency Measurement

Problem Statement: A stationary closed Lissajous pattern on an oscilloscope has 33 horizontal tangencies and 22 vertical tangencies for a horizontal input with frequency 3 kHz. Find out the frequency of the vertical input.

Answer

When the CRO is used in **X-Y mode**, the internal time-base is switched off. One signal is applied to the horizontal plates and the other signal is applied to the vertical plates. If the two signal frequencies are in a rational ratio, a stationary closed Lissajous figure appears on the CRO screen.

For Lissajous frequency measurement, the rule is:

$$\frac{f_y}{f_x} = \frac{\text{number of horizontal tangencies}}{\text{number of vertical tangencies}}.$$

Here,

$$f_x = 3 \text{ kHz}, \quad N_H = 33, \quad N_V = 22.$$

Therefore,

$$\frac{f_y}{3 \text{ kHz}} = \frac{33}{22} = \frac{3}{2}.$$

Hence,

$$f_y = 3 \times \frac{3}{2} \text{ kHz} = 4.5 \text{ kHz}.$$

$$\boxed{f_y = 4.5 \text{ kHz}}$$

Clean Rule Diagram

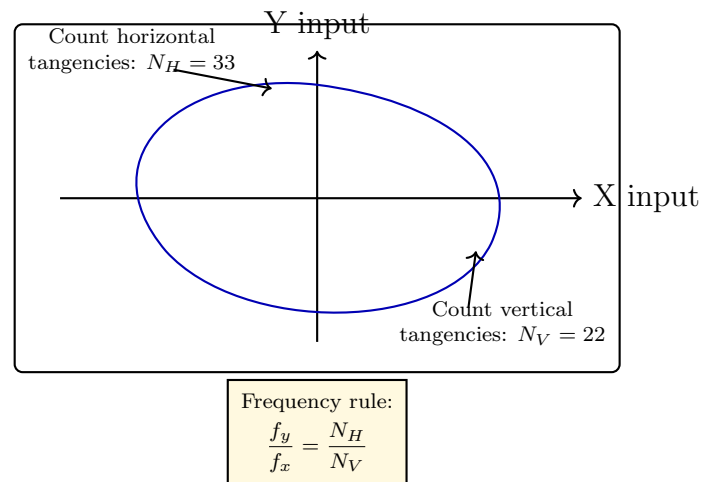


Figure 1: Lissajous tangency-count method for frequency measurement.

Final Exam Statement

Since the pattern has 33 horizontal tangencies and 22 vertical tangencies, the vertical-to-horizontal frequency ratio is $33 : 22 = 3 : 2$. For a horizontal input frequency of 3 kHz, the vertical input frequency is 4.5 kHz.

2 Solution to Question 1(b): Matched CRO Probe

Problem Statement: The input resistance and capacitance of a Cathode Ray Oscilloscope are $2.0\text{ M}\Omega$ and $1.0\text{ }\mu\text{F}$. What is the matched capacitance of the probe for proper compensation assuming $10\text{ k}\Omega$ probe resistance? Explain necessary formula.

Basic Idea

A CRO input is not purely resistive. It has an input resistance R_s and an input capacitance C_s . The probe also has a resistance R_p and an adjustable capacitance C_p . These two sections form a frequency-dependent voltage divider.

For undistorted waveform display, the attenuation should not depend on frequency. This is obtained when the time constant of the probe section equals the time constant of the CRO input section:

$$R_p C_p = R_s C_s.$$

Properly Spaced Equivalent Circuit

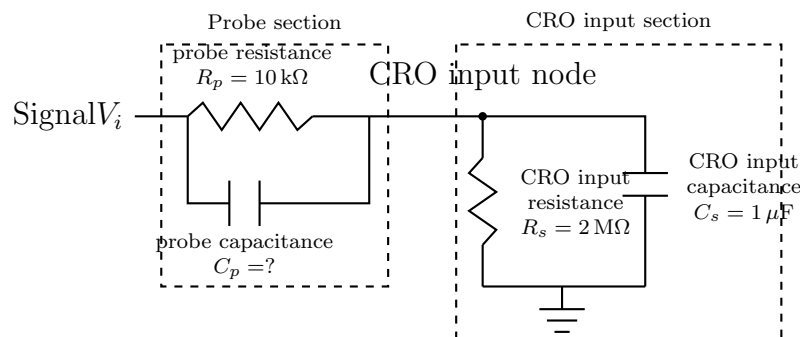


Figure 2: Compensated CRO probe and oscilloscope input equivalent circuit.

Derivation of Compensation Formula

For proper compensation, the capacitive and resistive divisions must give the same attenuation. The condition may be written as

$$R_p C_p = R_s C_s.$$

Therefore,

$$C_p = \frac{R_s C_s}{R_p}.$$

Substitute the given values:

$$R_s = 2.0 \times 10^6 \Omega, \quad C_s = 1.0 \times 10^{-6} \text{ F}, \quad R_p = 10 \times 10^3 \Omega.$$

Thus,

$$C_p = \frac{(2.0 \times 10^6)(1.0 \times 10^{-6})}{10 \times 10^3} = \frac{2.0}{10^4} = 2.0 \times 10^{-4} \text{ F}.$$

$$2.0 \times 10^{-4} \text{ F} = 200 \times 10^{-6} \text{ F} = 200 \mu\text{F}.$$

$$\boxed{C_p = 200 \mu\text{F}}$$

3 Solution to Question 1(c): Function of Delay Line in CRO

Problem Statement: Explain the function of a delay line in the vertical deflection system of a CRO.

Why Delay Line is Required

In a CRO, a portion of the input signal is taken to the trigger circuit. The trigger circuit starts the horizontal time-base generator. But the trigger circuit and time-base circuit require a small finite time to respond. If the vertical signal reaches the vertical plates immediately, the beginning of the waveform may already be over before the horizontal sweep starts.

Therefore a **delay line** is inserted in the vertical channel. It delays the vertical signal for a small time so that the horizontal sweep starts first and the complete leading edge of the waveform can be displayed.

Block Diagram

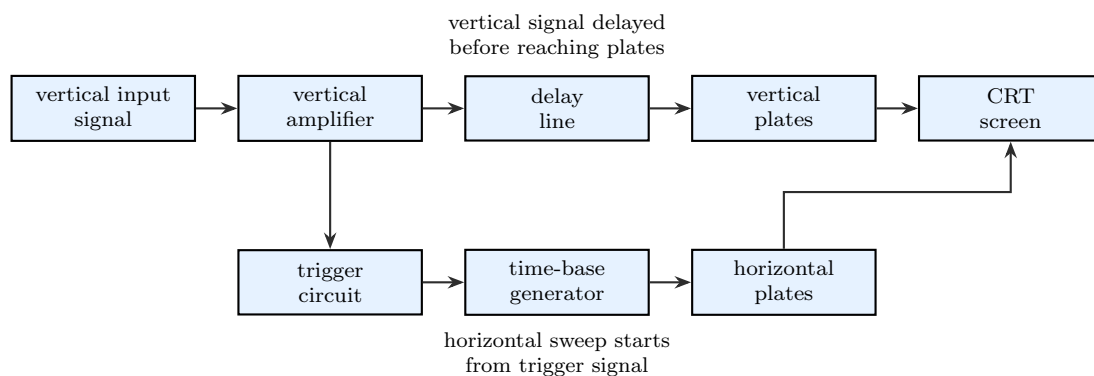


Figure 3: Delay line in the vertical channel of CRO.

Final Answer

The delay line delays the vertical input signal so that the time-base sweep can start before the signal reaches the vertical plates. Hence, the leading edge of a fast waveform is visible on the CRO screen.

For a short 2-mark answer: The delay line compensates for trigger/time-base starting delay and allows the complete waveform, especially the leading edge, to be displayed.

4 Solution to Question 1(OR)(a): X-Y Mode for Two In-Phase Sinusoids

Problem Statement: Two in-phase, 50 Hz sinusoidal waveforms of unit amplitude are fed into channel-1 and channel-2 respectively of an oscilloscope. Assuming that the voltage scale and time scale are same for both channels, what would be observed if the oscilloscope is operated in X-Y mode and why?

Physical Setup

In X-Y mode, the internal time-base is switched off. Channel-1 is applied to the X-plates and channel-2 is applied to the Y-plates. Since both signals are in phase, have the same amplitude, and have the same frequency, their instantaneous values are always equal.

Let

$$x = \sin \omega t, \quad y = \sin \omega t.$$

Therefore,

$$y = x.$$

This is the equation of a straight line passing through the origin with positive slope.

Expected CRO Pattern

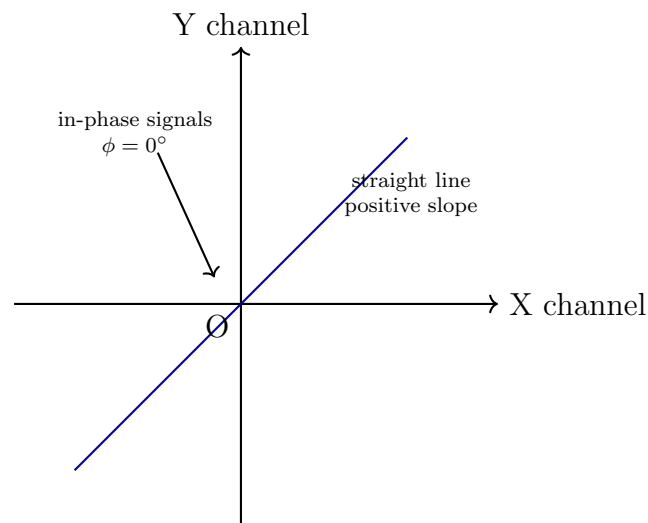


Figure 4: X-Y display for two equal-amplitude in-phase sinusoidal signals.

Conclusion

The CRO screen shows a straight line inclined at 45° to the X-axis if the two channel sensitivities are exactly equal. The line has positive slope because the two signals are in phase.

Observed pattern: straight line through origin with positive slope.

5 Solution to Question 1(OR)(b): Time-Base Generator Circuit in CRO

Problem Statement: Explain with circuit diagram the time-base generator circuit of a CRO.

Why Time Base is Required

The vertical plates show voltage variation. To show voltage as a function of time, the horizontal plates must move the beam at uniform speed from left to right. This requires a saw-tooth voltage:

$$V_x(t) = kt.$$

During the linear rise, the spot moves uniformly. During the rapid fall, it returns quickly to the starting point. This rapid return is called flyback or retrace.

Saw-Tooth Waveform

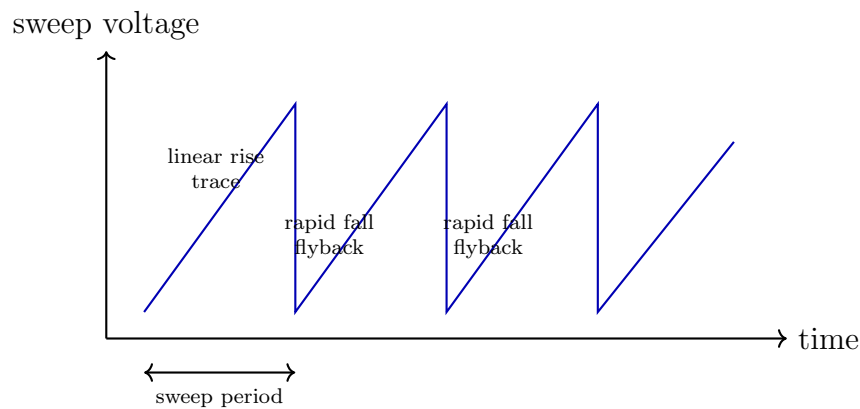


Figure 5: Saw-tooth voltage used as horizontal sweep voltage.

Basic RC Sweep Generator

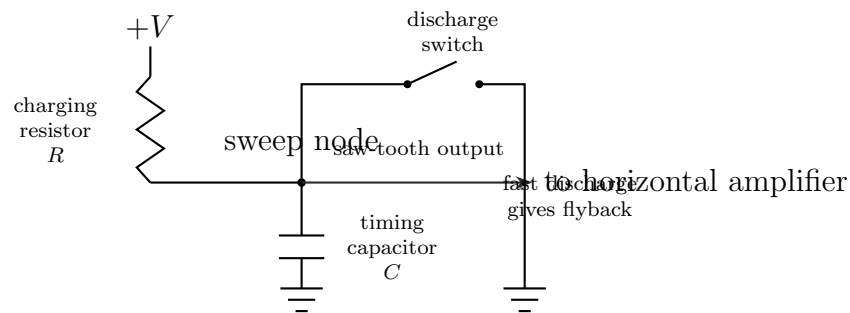


Figure 6: Basic time-base generator using capacitor charging and rapid discharge.

Operation

1. When the switch is open, capacitor C charges through R . Over the useful interval, its voltage rises almost linearly.
2. This rising voltage is applied to the horizontal plates, so the electron beam sweeps from left to right uniformly.
3. At the end of the sweep, the switch closes momentarily. The capacitor discharges rapidly.
4. The spot returns to the left side quickly. The process repeats and produces a periodic saw-tooth wave.

6 Solution to Question 2(a): State-Variable Filter as Universal Active Filter

Problem Statement: Why is a state-variable filter known as universal active filter? Draw the circuit diagram of a normalized band-stop state-variable filter and find out the transfer function, passband gain and bandwidth of the said filter.

Why It is Called Universal

A state-variable filter is built using a summer and two integrators. The same circuit simultaneously produces high-pass, band-pass and low-pass outputs. By taking suitable combinations of these outputs, one can obtain band-stop or notch response. Hence the same active circuit gives several standard filter responses, so it is called a **universal active filter**.

General Second-Order Denominator

The common denominator of all second-order state-variable outputs is

$$D(s) = s^2 + \frac{\omega_0}{Q}s + \omega_0^2.$$

The three basic outputs are

$$H_{HP}(s) = \frac{s^2}{D(s)}, \quad H_{BP}(s) = \frac{(\omega_0/Q)s}{D(s)}, \quad H_{LP}(s) = \frac{\omega_0^2}{D(s)}.$$

A band-stop output is obtained by adding high-pass and low-pass outputs:

$$H_{BS}(s) = H_{HP}(s) + H_{LP}(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

Normalized Band-Stop Case

For normalized response,

$$\omega_0 = 1 \text{ rad/s}.$$

Hence,

$$H_{BS}(s) = \frac{s^2 + 1}{s^2 + \frac{1}{Q}s + 1}.$$

The passband gain at very low frequency is

$$H_{BS}(0) = \frac{1}{1} = 1.$$

At very high frequency,

$$H_{BS}(\infty) = 1.$$

Therefore passband gain is

$$K = 1 = 0 \text{ dB}.$$

Bandwidth of a second-order band-stop or band-pass response is

$$B = \frac{\omega_0}{Q}.$$

For normalized $\omega_0 = 1$,

$$B = \frac{1}{Q}.$$

State-Variable Band-Stop Block Diagram

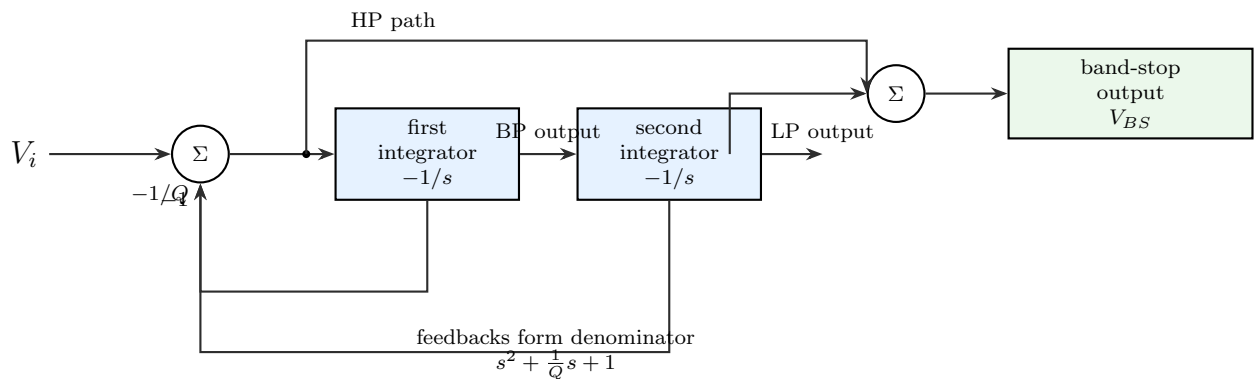


Figure 7: Normalized state-variable band-stop filter obtained by adding HP and LP outputs.

Final Result

$$H_{BS}(s) = \frac{s^2 + 1}{s^2 + \frac{1}{Q}s + 1}, \quad K = 1, \quad B = \frac{1}{Q}.$$

7 Solution to Question 2(OR)(a): PLL Stages and Linear Model

Problem Statement: What are the different stages of PLL? Explain the functions of each stage. Draw the block diagram of the linear model of PLL and hence deduce the transfer function of the same.

Definition

A phase-locked loop is a closed-loop feedback system that locks the phase and frequency of a voltage-controlled oscillator output to the phase and frequency of an input reference signal.

Stages of PLL

1. **Phase detector:** compares input phase θ_i and feedback phase θ_o , producing an error voltage proportional to phase error.
2. **Loop filter:** removes high-frequency components from the detector output and produces a smooth control voltage.
3. **Voltage-controlled oscillator:** generates an output frequency controlled by the loop filter voltage.
4. **Feedback path:** feeds the VCO output phase back to the phase detector. In frequency synthesizers a divider may be inserted here.

Linear PLL Model

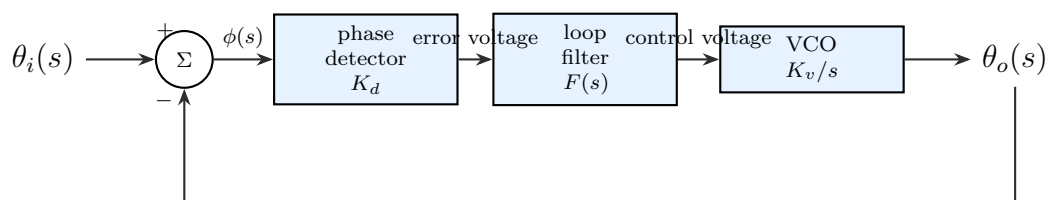


Figure 8: Linear model of a phase-locked loop.

Derivation of Closed-Loop Transfer Function

The phase error is

$$\phi(s) = \theta_i(s) - \theta_o(s).$$

Phase detector output:

$$V_e(s) = K_d[\theta_i(s) - \theta_o(s)].$$

Loop filter output:

$$V_c(s) = F(s)V_e(s) = K_dF(s)[\theta_i(s) - \theta_o(s)].$$

The VCO acts as an integrator because frequency is derivative of phase:

$$\theta_o(s) = \frac{K_v}{s}V_c(s).$$

Substituting,

$$\theta_o(s) = \frac{K_dK_vF(s)}{s}[\theta_i(s) - \theta_o(s)].$$

Let $K = K_dK_v$. Then

$$s\theta_o(s) = KF(s)\theta_i(s) - KF(s)\theta_o(s).$$

$$\theta_o(s)[s + KF(s)] = KF(s)\theta_i(s).$$

Therefore,

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}.$$

The phase error transfer function is

$$\boxed{\frac{\phi(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}}.$$

8 Solution to Question 2(b): PLL Frequency Synthesizer

Problem Statement: The block diagram of a frequency synthesizer consisting of a PLL and a divide-by- N counter with $/2, /4, /8, /16$ outputs is shown. The PLL input reference is 5 kHz. The free-running frequency is 20 kHz. A commutator switch repeatedly makes contact in the order 1–2–3–4. Find the synthesized frequencies.

Principle

In a PLL frequency synthesizer, the output of the VCO is divided by N and compared with the reference frequency. In locked condition,

$$\frac{f_o}{N} = f_{ref}.$$

Therefore,

$$f_o = N f_{ref}.$$

Here the available divider ratios are

$$N = 2, 4, 8, 16.$$

The reference frequency is

$$f_{ref} = 5 \text{ kHz}.$$

Synthesizer Diagram

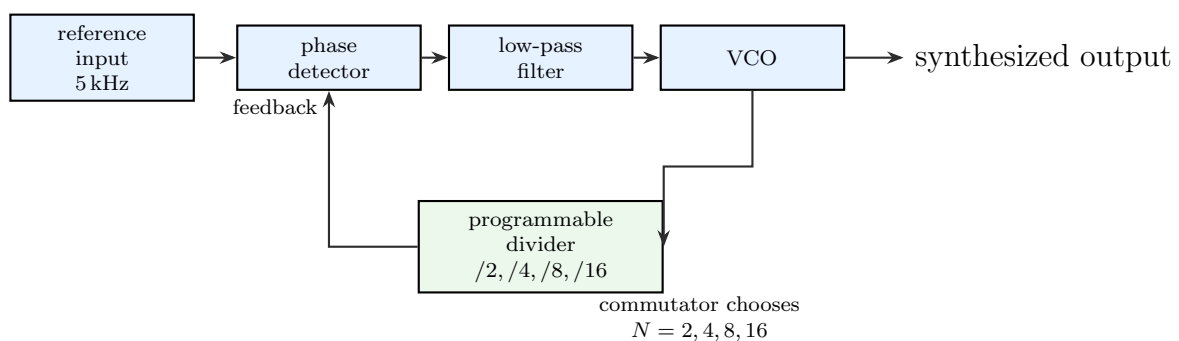


Figure 9: PLL synthesizer with selectable divider ratio.

Calculation

For contact 1, take $N = 2$:

$$f_o = 2 \times 5 \text{ kHz} = 10 \text{ kHz}.$$

For contact 2, take $N = 4$:

$$f_o = 4 \times 5 \text{ kHz} = 20 \text{ kHz}.$$

For contact 3, take $N = 8$:

$$f_o = 8 \times 5 \text{ kHz} = 40 \text{ kHz}.$$

For contact 4, take $N = 16$:

$$f_o = 16 \times 5 \text{ kHz} = 80 \text{ kHz}.$$

Final Answer

The commutator repeatedly selects the sequence

$$\boxed{10 \text{ kHz}, 20 \text{ kHz}, 40 \text{ kHz}, 80 \text{ kHz}}.$$

The free-running frequency 20 kHz indicates the natural VCO frequency before locking; the locked output is set by Nf_{ref} .

9 Solution to Question 3(a): 8-bit DAC Output Range with Full-Scale Error

Problem Statement: An 8-bit DAC has a full-scale output voltage of 2 V and a full-scale error of $\pm 5\%$ of full-scale output. What is the range of possible outputs for input 10000000?

Basic Meaning

An 8-bit DAC has $2^8 = 256$ possible input codes. The input 10000000₂ represents the decimal number

$$D = 1 \times 2^7 = 128.$$

This is exactly the mid-scale code in the common exam convention.

If the full-scale output is 2 V, then the ideal output for mid-scale is

$$V_{ideal} \approx \frac{128}{256} \times 2 = 1.0 \text{ V}.$$

The full-scale error is

$$\pm 5\% \text{ of } 2 \text{ V} = \pm 0.05 \times 2 = \pm 0.1 \text{ V}.$$

Therefore, the output range is

$$V_{min} = 1.0 - 0.1 = 0.9 \text{ V},$$

$$V_{max} = 1.0 + 0.1 = 1.1 \text{ V}.$$

$$0.9 \text{ V} \leq V_o \leq 1.1 \text{ V}$$

DAC Block Diagram

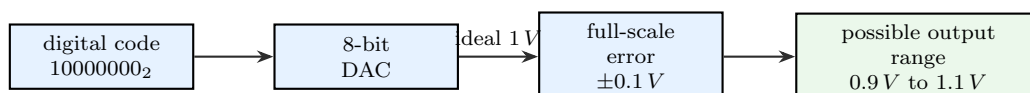


Figure 10: Block interpretation of DAC output range with full-scale error.

Note on Alternative Convention

If the full-scale output is strictly assigned to code $11111111_2 = 255$, then

$$V_{ideal} = \frac{128}{255} \times 2 = 1.0039 \text{ V},$$

and the range becomes approximately 0.9039 V to 1.1039 V. In most semester solutions, the simpler mid-scale convention gives 0.9 V to 1.1 V.

10 Solution to Question 3(b): Linearity Error of DAC

Problem Statement: Explain linearity error of DAC with the help of input/output relationship. Why does this type of error occur in DAC?

Ideal DAC Transfer Characteristic

For an ideal DAC, equal increments in digital input code produce equal increments in analog output voltage. Therefore, the input-output points lie on a straight line. If q is one LSB step size, then

$$V_o(D) = Dq + V_{offset}.$$

For an ideal zero-offset DAC,

$$V_o(D) = Dq.$$

Definition of Linearity Error

Linearity error is the maximum deviation of the actual DAC output from the ideal straight-line transfer characteristic after offset and gain errors have been removed.

$$\boxed{\text{Linearity error} = V_{actual} - V_{ideal\ line}}.$$

It is normally expressed as a fraction of LSB.

Diagram of DAC Linearity Error

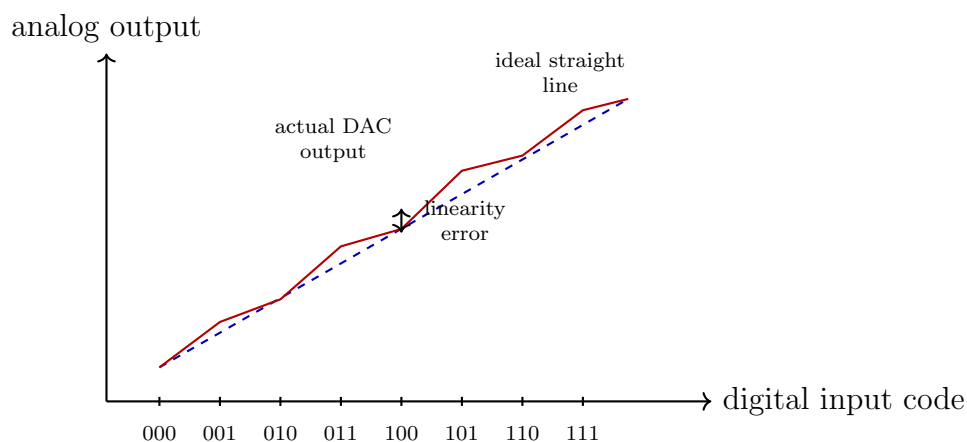


Figure 11: Linearity error of a DAC as deviation from ideal transfer line.

Why It Occurs

DAC linearity error occurs due to:

1. resistor mismatch in weighted resistor or R-2R ladder network;
2. switch ON resistance and leakage;
3. op-amp offset and finite gain;
4. reference voltage error;
5. temperature variation of circuit elements.

Final Answer

Linearity error represents the departure of actual DAC output from the ideal straight-line input-output relation. It occurs because practical circuit elements do not have exactly ideal binary weights.

11 Solution to Question 3(OR)(b): Dual-Slope Integrating ADC

Problem Statement: Explain the operation of a dual-slope integrator type Analog to Digital Converter.

Basic Principle

A dual-slope ADC converts an analog voltage into a time interval. It has two main phases:

1. **Run-up or integration phase:** the unknown input V_{in} is integrated for a fixed time T_1 .
2. **Run-down or de-integration phase:** a known reference voltage of opposite polarity is applied until the integrator output returns to zero.

Block Diagram

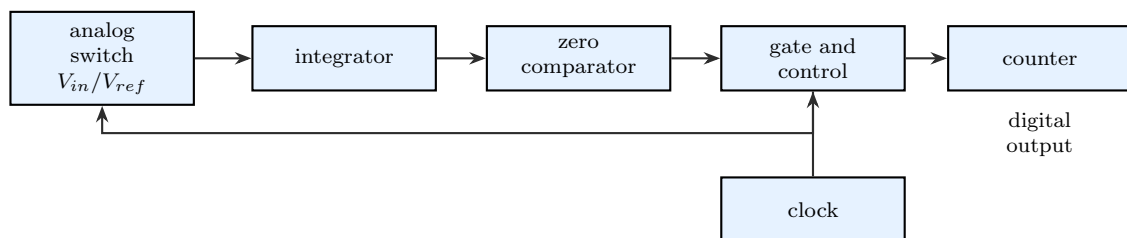


Figure 12: Block diagram of dual-slope ADC.

Mathematical Derivation

During fixed integration time T_1 , the integrator output becomes

$$V_o(T_1) = -\frac{1}{RC} \int_0^{T_1} V_{in} dt = -\frac{V_{in}T_1}{RC}.$$

During de-integration, a reference voltage V_{ref} of opposite polarity is applied for time T_2 . The change needed to return to zero is

$$\frac{V_{ref}T_2}{RC} = \frac{V_{in}T_1}{RC}.$$

Cancel RC :

$$V_{ref}T_2 = V_{in}T_1.$$

Therefore,

$$T_2 = \frac{V_{in}}{V_{ref}} T_1.$$

If clock frequency is f_c , the counted pulses are

$$N = f_c T_2 = f_c \frac{V_{in}}{V_{ref}} T_1.$$

Thus the digital count is directly proportional to V_{in} .

Waveform

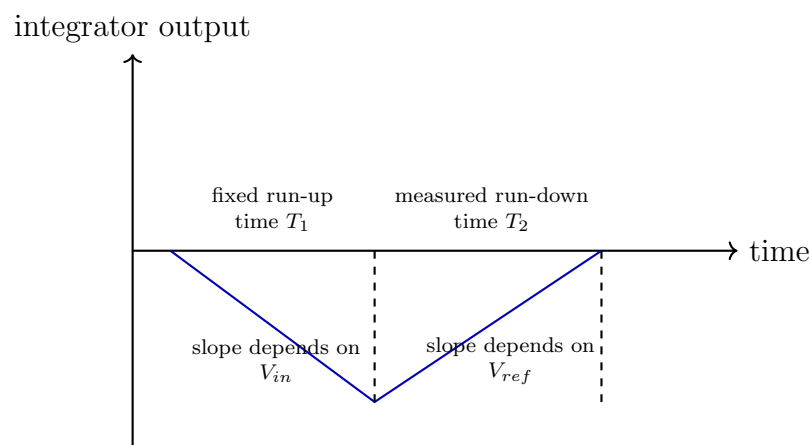


Figure 13: Integrator output waveform in dual-slope ADC.

Advantages

Dual-slope ADC has excellent noise rejection, especially for line-frequency noise, because integration averages the input over a fixed interval. It is highly accurate and is widely used in digital voltmeters.

12 Solution to Question 3(c): ADC Digital Code, Conversion Time and Quantization Error

Problem Statement: An 8-bit ADC has a full-scale output voltage of 10.23 V. The clock frequency is 10 MHz. Determine: (i) the digital equivalent corresponding to 3.72 V, (ii) conversion time and resolution for both counter-type and successive-approximation type converters. Also find quantization error and how it can be minimized.

Resolution

For an 8-bit ADC,

$$2^8 = 256 \text{ levels.}$$

Using the common endpoint convention,

$$q = \frac{V_{FS}}{2^8 - 1} = \frac{10.23}{255} = 0.04012 \text{ V.}$$

So

$$\boxed{q \approx 40.1 \text{ mV.}}$$

Digital Equivalent of 3.72 V

$$D = \frac{V_{in}}{q} = \frac{3.72}{0.04012} = 92.73.$$

Nearest integer code is

$$D \approx 93.$$

Convert 93 to binary:

$$93 = 64 + 16 + 8 + 4 + 1.$$

So,

$$93_{10} = 01011101_2.$$

$$\boxed{\text{Digital equivalent} = 01011101_2}$$

Conversion Time

Clock frequency:

$$f_c = 10 \text{ MHz} = 10 \times 10^6 \text{ Hz.}$$

Clock period:

$$T_c = \frac{1}{f_c} = 0.1 \mu s.$$

Counter-Type ADC

The counter counts from zero until DAC output reaches the input. For code 93,

$$T_{conv} = 93T_c = 93(0.1 \mu s) = 9.3 \mu s.$$

Worst-case time:

$$T_{max} = 255T_c = 25.5 \mu s.$$

Successive Approximation ADC

SAR ADC requires one clock cycle per bit. For 8 bits,

$$T_{conv} = 8T_c = 8(0.1 \mu s) = 0.8 \mu s.$$

Quantization Error

Maximum quantization error is half of one LSB:

$$E_q = \pm \frac{q}{2} = \pm \frac{40.12 \text{ mV}}{2} = \pm 20.06 \text{ mV}.$$

$$E_q = \pm 20.06 \text{ mV}$$

How to Minimize Quantization Error

Quantization error is minimized by reducing the step size. This is done by increasing the number of ADC bits. It can also be reduced effectively by oversampling, averaging and proper analog anti-alias filtering.

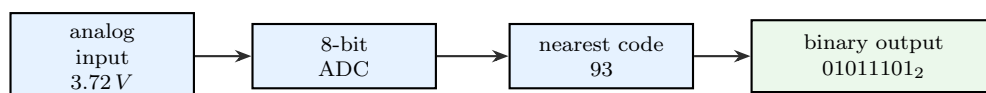


Figure 14: ADC conversion chain for 3.72 V.

13 Solution to Question 4(a): Butterworth Poles on Unit Circle

Problem Statement: Prove that the poles of a Butterworth filter are located on a unit circle.

Butterworth Magnitude Function

For a normalized n^{th} order Butterworth low-pass filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}.$$

The pole equation is obtained from

$$1 + (-1)^n s^{2n} = 0.$$

This equation has $2n$ roots equally spaced around a circle. For the stable filter we select only the roots in the left half of the s -plane.

The stable poles can be written as

$$s_k = -\sin \theta_k + j \cos \theta_k,$$

where

$$\theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, 2, \dots, n.$$

Now,

$$|s_k|^2 = (-\sin \theta_k)^2 + (\cos \theta_k)^2.$$

$$|s_k|^2 = \sin^2 \theta_k + \cos^2 \theta_k = 1.$$

Therefore,

$$\boxed{|s_k| = 1}.$$

Hence all normalized Butterworth poles lie on a unit circle.

Pole Diagram

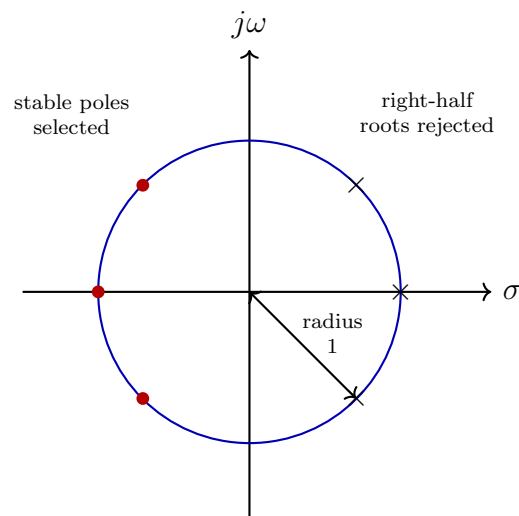


Figure 15: Normalized Butterworth poles lie on a unit circle; only left-half-plane poles are used.

14 Solution to Question 4(a)(OR): Chebyshev Properties and -3 dB Frequency

Problem Statement: Mention the properties of Chebyshev filter. Explain why the -3 dB cutoff frequency of a normalized Chebyshev filter is greater than 1 rad/s.

Properties of Chebyshev Filter

1. It has equiripple behaviour in the passband for Type-I Chebyshev filter.
2. It gives sharper transition than Butterworth filter of the same order.
3. Its passband ripple is controlled by ripple factor ϵ .
4. Its magnitude squared response is

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}.$$

5. Poles of Chebyshev filter lie on an ellipse in the s -plane.

Why -3 dB Frequency is Greater than 1 rad/s

For normalized Chebyshev low-pass filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}.$$

At $\omega = 1$,

$$T_n(1) = 1.$$

Therefore,

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}.$$

For usual passband ripple less than 3 dB, $\epsilon < 1$. Hence,

$$1 + \epsilon^2 < 2.$$

So,

$$\frac{1}{1 + \epsilon^2} > \frac{1}{2}.$$

But -3 dB means half-power:

$$|H(j\omega_{3dB})|^2 = \frac{1}{2}.$$

Therefore at $\omega = 1$, the response has not yet fallen to half power. Hence,

$$\omega_{3dB} > 1 \text{ rad/s}.$$

Response Diagram

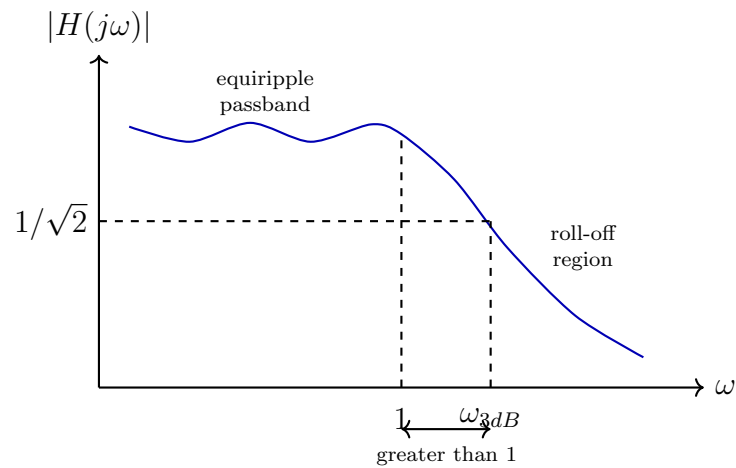


Figure 16: For a normalized Chebyshev filter, $\omega = 1$ is the end of ripple band, not necessarily the -3 dB point.

15 Solution to Question 4(b): Band-Pass and Band-Stop Design

Problem Statement: Find the transfer function of a band-pass filter circuit with centre frequency $\omega_0 = 10 \text{ krad/s}$ and bandwidth $B = 4 \text{ krad/s}$, obtained by low-pass to band-pass transformation from a first-order normalized low-pass Butterworth filter. Realize the band-pass circuit. Also realize the band-stop circuit from this band-pass filter having same centre frequency and passband gain 0 dB.

Low-Pass Prototype

The first-order normalized Butterworth low-pass prototype is

$$H_{LP}(s) = \frac{1}{s + 1}.$$

For low-pass to band-pass transformation,

$$s \rightarrow \frac{s^2 + \omega_0^2}{Bs}.$$

Therefore,

$$H_{BP}(s) = \frac{1}{\frac{s^2 + \omega_0^2}{Bs} + 1} = \frac{Bs}{s^2 + Bs + \omega_0^2}.$$

Given,

$$\omega_0 = 10 \text{ krad/s} = 10^4 \text{ rad/s},$$

$$B = 4 \text{ krad/s} = 4000 \text{ rad/s}.$$

Thus,

$$H_{BP}(s) = \frac{4000s}{s^2 + 4000s + 10^8}.$$

The quality factor is

$$Q = \frac{\omega_0}{B} = \frac{10^4}{4000} = 2.5.$$

Band-Pass Realization by State-Variable Structure

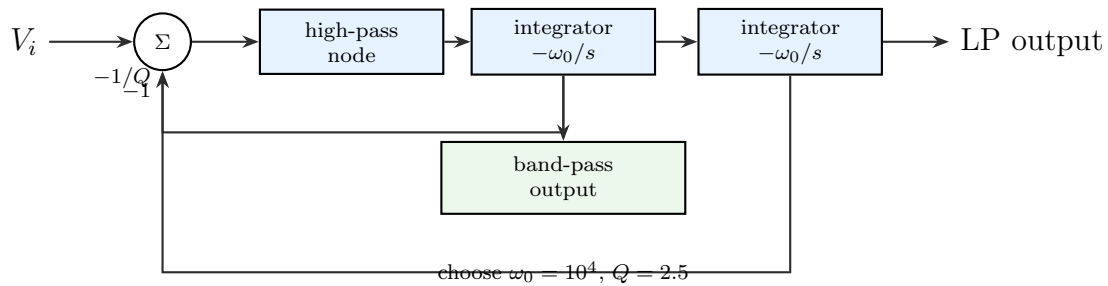


Figure 17: State-variable realization of the designed band-pass filter.

Band-Stop Transfer Function

For passband gain 0 dB, gain $K = 1$. A band-stop response can be formed as

$$H_{BS}(s) = 1 - H_{BP}(s).$$

Hence,

$$H_{BS}(s) = 1 - \frac{4000s}{s^2 + 4000s + 10^8} = \frac{s^2 + 10^8}{s^2 + 4000s + 10^8}.$$

$$H_{BS}(s) = \frac{s^2 + 10^8}{s^2 + 4000s + 10^8}.$$

At $s = j\omega_0 = j10^4$, numerator is $(j10^4)^2 + 10^8 = -10^8 + 10^8 = 0$. Hence the centre frequency is rejected.

Band-Stop Realization from Band-Pass Output

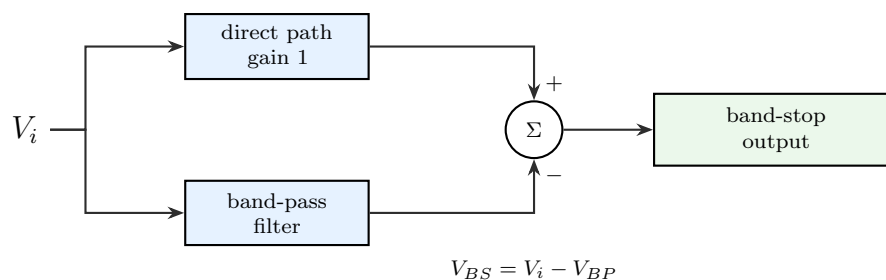


Figure 18: Band-stop filter obtained by subtracting band-pass output from direct signal.

16 Solution to Question 4(b)(OR): Normalized Butterworth LPF and Actual Realization

Problem Statement: Find the transfer function of a normalized Butterworth low-pass filter with specifications $|H(j0.5)|^2 > 0.9$ and $|H(j2)|^2 < 0.01$. Hence realize the actual low-pass filter circuit for cutoff frequency 1000 rad/s.

Butterworth Magnitude Squared Function

For normalized Butterworth response,

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}.$$

Passband Specification

At $\omega = 0.5$,

$$\frac{1}{1 + (0.5)^{2n}} > 0.9.$$

This gives

$$1 + (0.5)^{2n} < \frac{1}{0.9} = 1.1111.$$

$$(0.5)^{2n} < 0.1111.$$

This requires approximately

$$n > 1.58.$$

Stopband Specification

At $\omega = 2$,

$$\frac{1}{1 + 2^{2n}} < 0.01.$$

Thus,

$$1 + 2^{2n} > 100,$$

$$2^{2n} > 99.$$

This gives

$$n > 3.31.$$

Therefore the smallest integer order is

$$\boxed{n = 4}.$$

Normalized Fourth-Order Butterworth Transfer Function

The normalized fourth-order Butterworth denominator is

$$D(s) = s^4 + 2.6131s^3 + 3.4142s^2 + 2.6131s + 1.$$

Hence,

$$\boxed{H_n(s) = \frac{1}{s^4 + 2.6131s^3 + 3.4142s^2 + 2.6131s + 1}}.$$

It may be factored as

$$D(s) = (s^2 + 0.7654s + 1)(s^2 + 1.8478s + 1).$$

Frequency Scaling to $\omega_c = 1000 \text{ rad/s}$

Replace s by s/ω_c . Therefore,

$$H(s) = \frac{\omega_c^4}{s^4 + 2.6131\omega_c s^3 + 3.4142\omega_c^2 s^2 + 2.6131\omega_c^3 s + \omega_c^4}.$$

For $\omega_c = 1000$,

$$\boxed{H(s) = \frac{10^{12}}{s^4 + 2613.1s^3 + 3.4142 \times 10^6 s^2 + 2.6131 \times 10^9 s + 10^{12}}}.$$

Cascaded Realization

The fourth-order filter is realized by cascading two second-order low-pass sections:

$$H(s) = H_1(s)H_2(s).$$

The section quality factors are

$$Q_1 = \frac{1}{0.7654} = 1.3065, \quad Q_2 = \frac{1}{1.8478} = 0.5412.$$

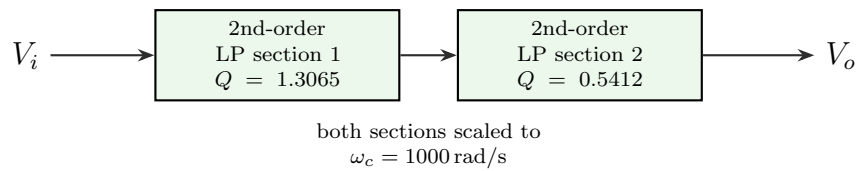


Figure 19: Actual fourth-order Butterworth low-pass filter realized as two cascaded second-order sections.

Final Answer

$$n = 4$$

$$H_n(s) = \frac{1}{s^4 + 2.6131s^3 + 3.4142s^2 + 2.6131s + 1}$$

$\omega_c = 1000 \text{ rad/s}$ scaled actual transfer function as given above.

Final Revision Summary

1. Lissajous frequency rule: $f_y/f_x = N_H/N_V$.
2. Matched CRO probe condition: $R_p C_p = R_s C_s$.
3. CRO delay line preserves the leading edge by delaying the vertical signal.
4. In-phase equal sinusoids in X-Y mode give a positive straight line.
5. Time-base generator produces linear saw-tooth voltage for horizontal sweep.
6. State-variable filter is universal because HP, BP, LP and BS outputs are obtainable from one structure.
7. PLL closed-loop transfer function: $\theta_o/\theta_i = KF(s)/(s + KF(s))$.
8. PLL synthesizer output: $f_o = N f_{ref}$.
9. DAC linearity error is deviation from ideal straight transfer line.
10. Dual-slope ADC converts voltage into run-down time and has strong noise rejection.
11. Butterworth poles lie on a unit circle.
12. Chebyshev -3 dB frequency is beyond 1 rad/s for ordinary ripple.
13. Band-pass from first-order LP prototype: $H_{BP} = Bs/(s^2 + Bs + \omega_0^2)$.
14. Fourth-order Butterworth satisfies the given normalized LPF specifications.